

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Period: \_\_\_\_\_

**Primary Objective**

By the end of this lesson, I will be able to...

1. Determine whether a rational function has a \_\_\_\_\_ or a \_\_\_\_\_ at a shared zero by comparing \_\_\_\_\_. (LO 1.10.A)
2. Find the \_\_\_\_\_ of a hole and express the result using \_\_\_\_\_. (LO 1.10.A)

**Vocabulary & Key Concepts****hole in a graph:**


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**removable discontinuity:**


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**common factor:**


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**limit of a function at a point:**


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**Hook**

Imagine your GPS loses signal for exactly one moment — at  $t = 5$  seconds — but your position function is defined everywhere else. The output at  $t = 5$  is missing, but by looking at outputs for inputs very close to 5, we can determine where the position function “should” be. This missing-but-predictable point is called a \_\_\_\_\_ in the graph of a rational function.

**Holes vs. Vertical Asymptotes**

Let  $r(x)$  be a rational function with a shared zero at  $x = a$ . Let  $m$  = multiplicity of  $a$  in the **numerator** and  $n$  = multiplicity of  $a$  in the **denominator**.

| Condition               | Feature at $x = a$ | Example                                 |
|-------------------------|--------------------|-----------------------------------------|
| $m \geq n$              | _____              | $\frac{(x-2)^2}{x-2}$ : hole at $x = 2$ |
| $m < n$                 | _____              | $\frac{x-2}{(x-2)^2}$ : VA at $x = 2$   |
| $a$ only in denominator | _____              | $\frac{5}{x-2}$ : VA at $x = 2$         |

**Key idea:** A hole occurs when the numerator and denominator *share* a zero *and* the numerator's multiplicity is \_\_\_\_\_ or \_\_\_\_\_ the denominator's.

### Finding the Coordinates of a Hole

#### 4-Step Procedure:

1. \_\_\_\_\_ the numerator and denominator.
2. Identify \_\_\_\_\_ zeros and compare multiplicities.
3. \_\_\_\_\_ the common factor from numerator and denominator.
4. Substitute  $x = c$  into the \_\_\_\_\_ form to find  $L$ .

**Example 1.**  $r(x) = \frac{x^2 - 4}{x - 2}$

Step 1 — Factor:  $r(x) = \frac{\text{_____}}{x - 2}$

Step 2 — Shared zero:  $x = \text{_____}$  Num. mult: \_\_\_\_\_ Denom. mult: \_\_\_\_\_ Since  $m$  \_\_\_\_\_  $n$ , this is a \_\_\_\_\_.

Step 3 — Cancel: simplified form = \_\_\_\_\_

Step 4 — Substitute  $x = 2$ :  $L = \text{_____}$

**Hole at:**  $(\text{_____, } \text{_____})$

### Limit Notation for Holes

If a rational function  $r$  has a hole at  $(c, L)$ , we write:

$$\lim_{x \rightarrow c} r(x) = L$$

This means: as the input  $x$  gets \_\_\_\_\_ to  $c$  (from either side), the output  $r(x)$  gets \_\_\_\_\_ to  $L$ .

Both one-sided limits agree:  $\lim_{x \rightarrow c^-} r(x) = \text{_____}$  and  $\lim_{x \rightarrow c^+} r(x) = \text{_____}$

**Example 2.**  $r(x) = \frac{x^2 - 1}{(x - 1)(x + 2)}$

Shared zero at  $x = 1$ . Simplified form: \_\_\_\_\_

$L = \text{_____} = \text{_____}$

Limit statement:  $\lim_{x \rightarrow 1} r(x) = \text{_____}$  Hole at:  $(\text{_____, } \text{_____})$

Is there a VA? \_\_\_\_\_ If so, where? \_\_\_\_\_